

Marketing Campaign Performance Analysis

Introduction

This project focuses on analysing marketing campaign data to evaluate performance across different campaign types, locations, and customer segments. The objective is to derive meaningful insights using data cleaning, SQL analysis, and visualization.

Tools used

Python – Data cleaning and preprocessing

SQL – Data querying and analysis

Microsoft Power BI Desktop – Dashboard and visualization

Dataset Overview

The dataset contains information related to marketing campaigns including:

- Campaign details (Campaign_ID, Company, Campaign_Type)
- Audience (Gender, Age_Group, Customer_Segment)
- Performance metrics (ROI, Conversion Rate, Clicks, Impressions)
- Cost metrics (Acquisition Cost, Cost per Click, Profit)
- Time-based data (Date, Month, Year)

Data Preprocessing (Python)

The dataset was cleaned and transformed using Python:

- Converted Date column to datetime format
- Extracted Year, Month, Month_Name
- Cleaned Acquisition Cost (removed symbols like \$, ,)

Created new metrics:

- Click Through Rate (CTR)
- Cost per Click (CPC)
- Cost per Impression

Handled missing values and ensured correct data types

SQL Analysis

SQL was used to extract insights:

- Campaign performance based on Average ROI
- Channel effectiveness across locations
- Customer segment analysis
- Monthly performance trends
- Cost vs ROI comparison

- Engagement and click through rate analysis
- Identification of high-performing campaigns

Dashboard (Power BI)

An interactive dashboard was created using Microsoft Power BI Desktop with 3 pages:

Page 1: Overview

- KPIs: Average ROI, Total Clicks, Total Impressions, CTR
- Campaign-wise and location-wise performance
- Customer segment distribution

Page 2: Performance Analysis

- Monthly ROI trends (line chart)
- Campaign-wise engagement and CTR
- Scatter plot (Cost vs ROI)

Page 3: Insights

- Top campaigns based on ROI table
- Map showing ROI in various locations
- Filters (Year, Location, Campaign Type) for interactivity

Key Insights

- ROI across campaign types is relatively consistent, indicating stable performance
- Some locations perform slightly better, showing geographical variation
- Customer segments show different engagement patterns
- Click through rate and Engagement provide deeper insights beyond ROI
- Cost vs ROI analysis helps identify efficient campaigns

Conclusion

This project demonstrates end-to-end data analysis:

- Data cleaning using Python
- Insight generation using SQL
- Visualization using Power BI

The dashboard provides a clear understanding of marketing performance and helps in making data-driven decisions.

Future Improvements

- Use real-world dataset with higher variation
- Add predictive analysis (machine learning)
- Integrate real-time dashboards